

Influence of Zr addition on precipitation evolution and performance of Al-Mg-Si alloy conductor

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ABSTRACT

The precipitation evolution of Al-Mg-Si-Zr alloy samples was observed by transmission electron microscopy. Addition of Zr into the Al-Mg-Si alloys significantly hindered the growth of the Mg/Si particles and had a positive effect on the particle density. The higher the Zr content, the more difficult was the diffusion of the Mg and Si atoms. The sample with 0.13 wt% Zr showed the best combination of electrical conductivity (34.44 MS/m), tensile strength (198 MPa), and elongation of 17.8%.

1. Introduction

Electrical power transmission systems play a critical role in the energy market. More than 40% of primary energy is transformed into electrical power and is transmitted to families and factories all over the world by various types of aluminum cables. There are two solutions to improve the efficiency of an electrical power transmission system. One is to increase the conductivity of the cables by adding beneficial elements, such as rare earth and boron, to improve the electrical properties [1–3], which can decrease the loss of electrical power. The other is to increase the capacity of the transmission system by elevating the voltage of electrical power, which has a lower relative power loss [4]. As the potential transformer technology has improved, the later solution has shown to be more efficient and has been gradually applied in Chinese 1 MV electrical transmission systems. However, the later solution requires overhead cables with a higher performance.

When the capability of the electrical transmission system is improved, more Joule heat and higher working temperatures (above 150 °C) are expected. Thermal-resistant aluminum conductor steel-reinforced (TACSR) cables made of Al–Zr alloy, which are characterized by their permissible work temperature in the range of 150–230 °C, are considered to be one of the most qualified candidates [5,6]. However, TACSR cables are not still as good as all aluminum alloy conductor (AAAC) cables because of its lower corrosion resistance [7]. The development of AAAC cables with a work temperature of 150 °C has become more promising in transmission lines.

An Al-0.13 wt% Zr alloy wire has a good electrical conductivity, however, its tensile strength is between 150 MPa and 180 MPa [8], which does not meet the qualifications of an AAAC cable. Al-Mg-Si

alloys, such as commercial AA6101/AA6201, are widely used as materials for AAAC cables for overhead transmission lines [9]. They have a good combination of strength and electrical conductivity; however, their permissible work temperatures are low.

In order to develop an AAAC cable with a work temperature of 150 °C and a conductivity of 34.8 MS/m, efforts have been made to improve the thermal-resistant properties of Al-Mg-Si alloy cables. The addition of Zr to Al-Mg-Si alloys is expected to improve strength and heat resistance. Bahrami [10] added 0.1% Zr into an Al-Mg-Si alloy. The strength and elongation values increased from 160 MPa and 3.2% to 292 MPa and 9.5%, respectively. Yuan et al. [11] demonstrated that addition of 0.145% Zr to 6101 alloys provided minimal softening under a thermal stability treatment (180 °C) for 400 h. Dang et al. [12] found that Al-Mg-Si-Mn alloys with 0.2%–0.4% Sc and 0.1%–0.2% Zr could refine the grain size and improve the mechanical properties.

However, the fundamental mechanism of the influence of the Zr additions on the Mg₂Si precipitates of the Al-Mg-Si alloy has not been discussed. For Al-Mg-Si alloys, a thermal treatment process forms second-phase precipitates of Mg₂Si through the conventional $\alpha(\text{Al}) \rightarrow \text{GP zone} \rightarrow \beta'' \rightarrow \beta' \rightarrow \text{Mg}_2\text{Si}$ phase transformation sequence [13–15]. Various Mg/Si ratios affect the microstructure and performance properties of the Al-Mg-Si alloy cables [16]. Zr diffuses slowly in α -Al, and the L1₂ structure of Al₃Zr is thermodynamically metastable, inhibiting recrystallization to insure strength during a high temperature [17]. At approximately 475 °C, the L1₂(Al₃Zr) precipitates coarsen and transform into their equilibrium D0₂₃ structure [18–20]. L. Litynska found that in the Al-1.0 Mg-0.6Si-0.5Zr alloy, the primary particles of the Al₃Zr phase could form aggregates with the Mg₂Si and AlFeSi phases at the grain boundaries [21].

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Table 1
Chemical composition of the experiment alloys.

Alloys	Mg/wt.%	Si/wt.%	Zr/wt.%	Al/wt.%
Without Zr	0.60	0.49	0	Bal.
Alloy 1	0.53	0.45	0.13	Bal.
Alloy 2	0.59	0.45	0.18	Bal.
Alloy 3	0.58	0.49	0.23	Bal.

In this study, the effects of the addition of Zr and various Zr contents on the microstructure and mechanical and electrical properties of Al-Mg-Si alloy cables were evaluated.

2. Experimental

Four experimental alloys (in Table 1) were prepared in a resistance furnace using industrial pure aluminum (99.7%). Al–20Si, Al–10Zr master alloy, Al–4B master alloy, and pure magnesium ingot were added in order into molten Al. The molten metal was deslagged and degassed by 5 N Argon air and held for 15 min. Thereafter, the melt was poured into a cylindrical copper mold with a cavity size of $\Phi 50$ mm $\times$ 100 mm. In order to reduce the content segregation, the ingots were homogenized at 530 °C for 14 h in a muffle resistance furnace after surface milling. Then, the ingots were extruded to $\Phi 10$ mm rods with the extrusion initial temperature of 460 °C and the extrusion ratio of 25. Finally, the rods were drawn with multi-passes ($\Phi 10$ mm– $\Phi 9$ mm– $\Phi 8$ mm) by a chain draw bench at room temperature. To evaluate the mechanical and electrical properties of the specimens during heat treatment, an artificial aging treatment was carried out in a muffle resistance furnace at 200 °C for 2, 4, 6, 12, 24, 36, 48, 72, and 96 h.

The tensile test was carried out on a CSS-4100 electronic universal testing machine with a load speed of 2 mm/min at room temperature. In addition, the electrical conductivity was measured by a Wheatstone bridge method at 20 °C (ZhengYang 9510 micro-Ohm meter). Measurements of the strength and electrical conductivity were obtained for each heat treatment cycle.

The tested alloy samples were electrochemically etched in a solution of 7 vol% HBF₄ and 93 vol% H₂O, and were observed under polarized light by OLYMPOUS-PMG3 optical microscope (OM). The change of morphology of the precipitates was investigated by TEM using a Tecnai G20ST Microscope operated at 200 kV. The TEM samples were prepared by twin jet chemical polishing in an electrolyte containing 30 vol % HNO₃ + 70 vol% CH₃OH at a temperature of –30 °C and voltage of 20 V. All the TEM analyses were performed along $\langle 001 \rangle_{\text{Al}}$ directions, where approximately 1/3 of the precipitate could be viewed in the cross section and 2/3 could be imaged perpendicular to the needle lengths [22]. The average precipitate needle-lengths and number densities were quantified from the bright-field TEM images. The average precipitate needle-lengths were calculated from three measurements of the average needle-lengths in three different observed regions that were not on the grain boundaries. The two dimensional number density of precipitate was defined as $N_A = N / A$, where N is the number of the needles in the zone-axis direction (the dark spots in Fig. 4), the parameter A is the respective area on the TEM image. According to the stereological method, the three dimensional number density can be calculated from the two dimensional number density: $N_V = 2N_A / 3L$, where L is the mean length of particles. Considering that there were three directions of the needles, the precipitate number density was expressed as $P_V = 3N_V = 2N / AL$.

3. Results

3.1. Electrical properties and mechanical properties

Fig. 1(a) shows the electrical conductivity curves of the alloys with

different Zr percentages. The conductivity of each alloy sample increased significantly with time and finally reached the maximum. The addition of the Zr element hindered the conductivity. The maximum conductivity of the alloy without Zr was 34.86 MS/m, while those of the alloy with 0.13, 0.18, and 0.23 wt% Zr were 34.44, 33.52, and 33.21 MS/m, respectively. The slope of the conductivity curves decreased as the Zr content increased. Thus, the electrical conductivity curve reached a plateau more slowly than that of the lower Zr content. The Zr-free alloy samples plateaued after approximately 24 h of aging time, while the Zr-containing samples plateaued after 48 h.

Fig. 1(b) shows the tensile strength evolution during aging. The tensile strength of the alloy without Zr increased rapidly to a sharp peak in the early stage of aging. After a slight decrease, the strength increased to a second peak with an increasing aging time. Upon further heating, the material was “over-aged”, and the curves showed a continuous decline. An obvious two tensile strength peaks occurred, as indicated by arrows A1 and A2. Fig. 1(b) shows that adding the Zr element increased the tensile strength of the alloys. However, the tensile strength was not linearly proportional to the Zr percentage. The alloy with 0.18 wt% Zr achieved the highest tensile strength, 230.5 MPa, and its elongation was 22.7%. A higher or lower Zr content resulted in a lower tensile strength, as shown in Table 2. Although the alloy with 0.13 wt% Zr achieved a lower tensile strength of 197 MPa and elongation of 17.8 wt%, it showed the best combination of properties. The tensile strength curves contained inconspicuous two tensile strength peaks under the experimental conditions. As the arrows A1-D1 and A2-D2 indicated in Fig. 1(b), the peaks of the alloy without Zr occurred earlier. The peaks of the alloy with a higher Zr wt% lagged behind the alloy with a lower Zr wt%. The higher the Zr wt%, the later the peaks appeared. It was thought that Zr addition affected the diffusion of Mg and Si atoms in Al matrix and changed the deposit behavior of Mg/Si particles.

3.2. Microstructure investigation

The optical micrographs of longitudinal section structure of the tested alloys with different Zr addition are presented in Fig. 2. It can be seen that there was a set of nearly equiaxed grains in the tested alloy bars after hot extrusion. But the grain size of alloy without Zr was significantly larger than that of the alloy containing Zr. When the Zr content reached 0.23 wt%, the grain size of as-extruded alloy was smaller. The grain sizes of the samples A to D in longitudinal direction are 182.9 μm , 110.8 μm , 116.6 μm , and 86.3 μm , respectively. The grain sizes in radial direction are 72.0 μm , 64.5 μm , 87 μm , and 60.7 μm , respectively. In the hot extrusion process, the Al alloy is prone to dynamic recrystallization [23]. It can be concluded that trace amount of Zr can make a significant contribution to hinder the migration of grain boundaries, to suppress the dynamic recrystallization. According to the Hall-Petch theory, metals with the grain size larger than 10 nm generally agree with classical Hall-Petch equation [24], as Eq. (1) demonstrates,

$$\sigma_y = \sigma_0 + k \cdot d^{-1/2}, \quad (1)$$

where σ_y is yield strength, σ_0 and k are constants, d is the average grain size for Al-Mg-Si-Zr alloy. The finer alloy grains are more favorable for improving the strength properties of alloys. However, it can be found in Figs. 1 and 2, as the aging time increased, the relationship between tensile strength and grain size gradually deviated from Hall-Petch equation. The grain size of the alloy with 0.23 wt% Zr was smaller than the alloy with 0.18 wt% Zr; however, the tensile strength of the alloy with 0.23 wt% Zr was lower than the alloy with 0.18 wt% Zr. It can be speculated that Zr addition might also cause other effects.

In order to clarify whether Zr had an influence on the Mg/Si particle nucleation and growth, the microstructure of samples aged at 200 °C for different times was investigated. It can be seen in Fig. 3 that the

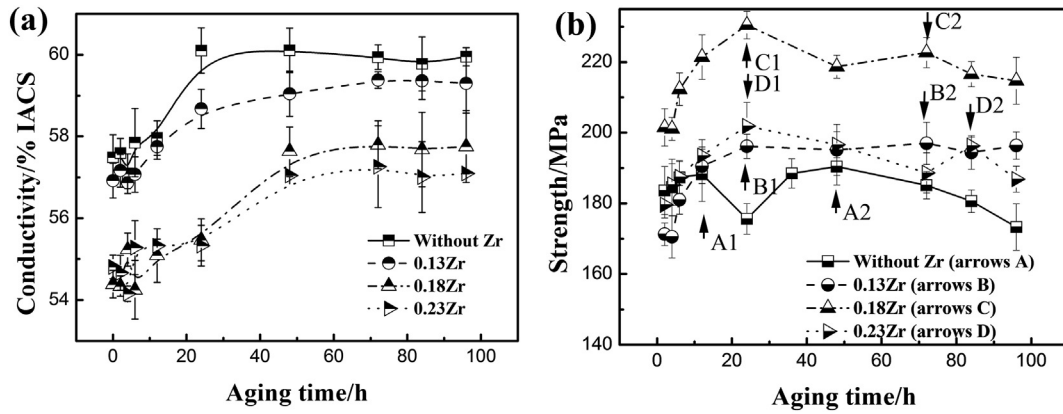


Fig. 1. Relationships between the performance of the alloy samples and the aging time for the alloys with various Zr content: (a) electrical conductivity and (b) tensile strength.

precipitation processes of the Mg/Si particle in alloy samples with different Zr contents showed obviously difference. A delay phenomenon of the evolution of the Mg/Si particle caused by Zr can be seen visually in Fig. 3. This corresponded to the results in Fig. 1 that Zr delayed the inflexion of the electrical conductivity curve and the valley of the strength curve. The electrical conductivity and aging strengthening is related to Mg/Si particle deposition, and Zr is thought to play an important role in Mg/Si particle deposition.

3.3. Quantitative measurements of the precipitates

Fig. 4 shows the evolution of the Mg/Si needle particles as a function of time. In Fig. 4(a), when the aging time prolonged, the Mg/Si particles grow longer. Fig. 4(a) also shows that when the Zr percentage was greater, the particles size was smaller. Zr played an important role in the growth of the Mg/Si particles and hindered the growth of the Mg/Si particles. Fig. 4(b) shows the evolution of the particles density of the Mg/Si particles as the aging time increased. It can be seen that the Zr-containing alloys had more Mg/Si particles than the alloys without Zr. The density of Mg/Si particles in the samples without Zr decreased quickly with an increasing aging time, and after 48 h, the Mg/Si particles was almost invisible, which means that most of the β'' phase needles were converted into coarser β' and β phases, as shown in Fig. 3(c). In the case of the Zr contained alloys, the density of Mg/Si particles still kept a high value after 96 h aging. This may be due to the fact that the addition of Zr hindered diffusion of Mg and Si atoms and delayed the Oswald ripening process.

4. Discussion

The diffusion rate of Zr in the Al matrix was low. Therefore, it was difficult to achieve good precipitation hardening at low aging temperatures and in short aging times. When Al_3Zr particles precipitate, the

cables can show good heat resistance because Al_3Zr particles are stable at elevated temperatures [25]. The drawbacks of Al–Zr cables are high cost and relatively low strength [26]. When a small amount of Zr was added to the Al–Mg–Si alloy, Zr did not precipitate on a large scale, and Al_3Zr particles were not the hardening phase. However, a small amount of Zr did improve the mechanical properties and heat resistance of the Al–Mg–Si alloy.

The intracrystalline diffusion is controlled by di-vacancies migration and vacancy interactions with solute atoms [27]. Thus, vacancies are critical for nucleation and growth of precipitated phases. S. Pogatscher et al. thought that artificial aging of Al–Mg–Si alloys is mainly controlled by the concentration of mobile vacancies [28]. According to Eui Pyo Kwon et al. [29], Zr preferentially combines with vacancies because Zr (0.33 eV) has a higher vacancy binding energy than that of Mg atoms (0.15 eV). This indicates that the higher is zirconium content, the lower is the concentration of mobile vacancies. The intracrystalline diffusion of Mg and Si atoms through vacancies may be hindered by Zr. The diffusion can be accelerated by dislocations [30]. The atomic sizes of Al, Mg, Si and Zr are 0.143 nm, 0.160 nm 0.118 nm and 0.160 nm, respectively [31]. Both Mg atoms and Zr atoms are bigger than Al atoms, and Zr atoms and Mg atoms tend to aggregate at the bottom of the partial atomic plane of the edge dislocations. For the diffusion rate of Zr in crystal defects is very small [17], the diffusion rate of Mg atoms will be hindered by Zr. It is noted that only several Zr atoms can block a whole dislocation. Therefore, trace amounts of Zr can block the diffusion along the dislocations.

In the early aging, the diffusion rate of Mg is slowed down by Zr atoms intracrystallinely as well along the dislocation. Mg–Si co-clusters have less chance to reach the critical nucleation radius. Thereby, it can be seen in Fig. 3 and Fig. 4 that the increasing content of Zr prolonged the incubation period of Mg/Si particles. Trace amounts of zirconium, which evenly distribute in the matrix, mainly influence the long-range diffusion of the Mg/Si atoms. Therefore, the growth and evolution of

Table 2

The mechanical properties of the samples aged for different times.

Aging time/h	0 Zr		0.13Zr		0.18 Zr		0.23Zr	
	σ_b /MPa	δ /%	σ_b /MPa	δ /%	σ_b /MPa	δ /%	σ_b /MPa	δ /%
2	184 ± 7	10.4 ± 2.1	171 ± 3	15.5 ± 1.8	201 ± 5	10.3 ± 2.8	180 ± 4	16.6 ± 4.1
4	185 ± 8	8.6 ± 3.4	171 ± 6	15.6 ± 1.2	203 ± 3	17.9 ± 2.2	185 ± 6	18.1 ± 2.4
6	187 ± 4	11.2 ± 2.2	181 ± 4	16.4 ± 2.3	212 ± 5	17.7 ± 1.1	187 ± 5	18.9 ± 2.8
12	188 ± 8	10.5 ± 2.4	190 ± 5	16.8 ± 1.5	221 ± 6	20.0 ± 1.9	194 ± 4	19.1 ± 2.4
24	176 ± 4	10.9 ± 2.1	196 ± 3	17.8 ± 2.1	230 ± 4	22.7 ± 2.1	202 ± 7	16.2 ± 2.0
48	190 ± 5	13.6 ± 3.2	195 ± 7	19.5 ± 1.7	219 ± 3	18.6 ± 1.3	197 ± 4	15.4 ± 1.8
72	185 ± 4	14.2 ± 2.9	197 ± 6	19.3 ± 1.8	223 ± 4	20.6 ± 2.1	189 ± 6	16.5 ± 3.5
96	173 ± 7	12.7 ± 1.2	196 ± 4	17.4 ± 2.5	215 ± 7	17.8 ± 3.1	187 ± 4	16.8 ± 1.9

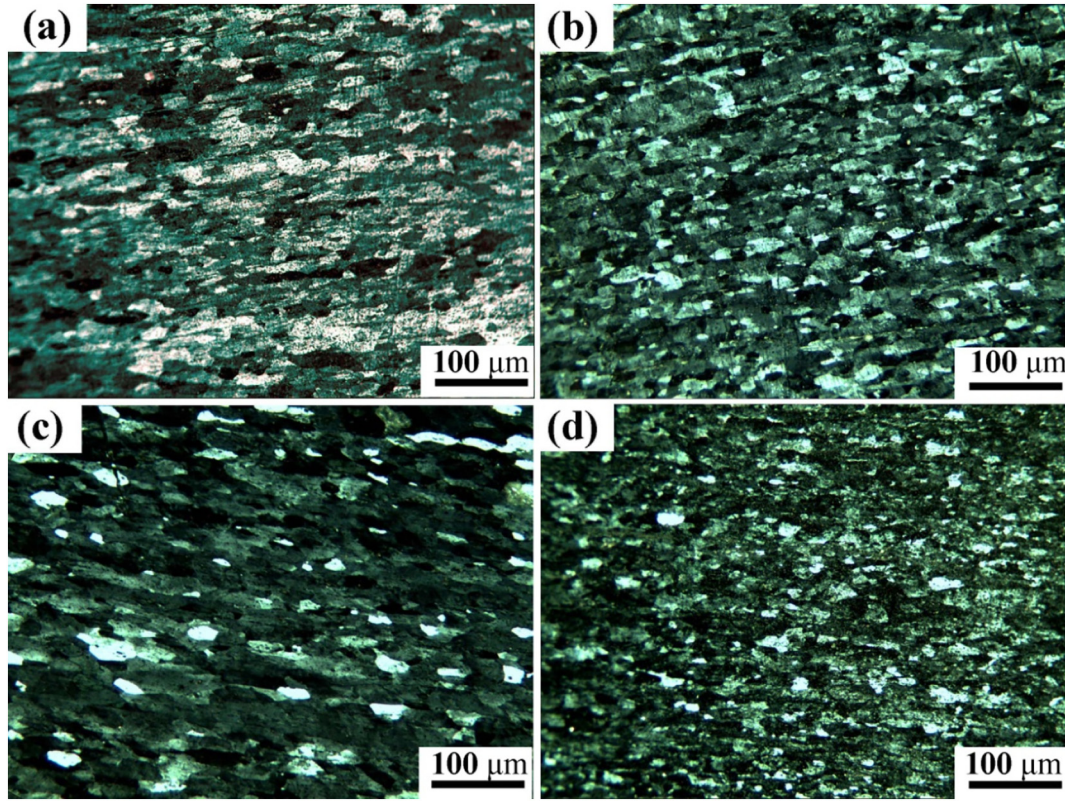


Fig. 2. Optical micrographs of the longitudinal section structure after hot extrusion: (a) Without Zr (b) Alloy 1 (Zr = 0.13 wt%) (c) Alloy 2 (Zr = 0.18 wt%) (d) Alloy 3 (Zr = 0.23 wt%).

the Mg/Si particles are retarded due to the slower long-range diffusion rate caused by Zr. The higher the Zr content, the shorter the Mg/Si particles.

According to S. Pogatscher [28], the Si–Mg co-clusters provide nucleation sites for the subsequent precipitates. In the case of no zirconium addition, the Mg/Si new phases grow rapidly at elevated temperatures. When they grow up, they absorb a large number of solute atoms. This will dissolve the small Si–Mg co-clusters nearby before they reach the critical nucleation radius. When Zr is added, Zr content will reduce the vacancy concentration and block the diffusion path along the dislocations. It is thought that the growth of primary precipitated particles will be delayed. This delay gives many small Si–Mg co-clusters enough chance and time to develop into nuclei. Therefore, the density of nuclei in Zr-containing alloy is much higher than that of the alloy without Zr. And the alloy with 0.18% Zr content has the densest Mg/Si particles. It is thought that Zr content higher than 0.18% hinder not only long range diffusion, but also short range diffusion. Therefore, both nucleation and growth of Mg/Si particles are hindered. The density of Mg/Si particles slightly decreased.

To further clarify the obstruction to the Mg/Si particles evolution, the apparent diffusion coefficient of the Mg/Si atoms was calculated. The Ostwald ripening model was used to describe the kinetics of the growth of the precipitates as follows [11,32].

$$\frac{dr}{dt} = \frac{2DV_m C_\alpha(\infty)}{RT} \frac{1}{r} \left(\frac{1}{r_0} - \frac{1}{r} \right). \quad (2)$$

The parameter r is the equivalent radius at time t , and D is the apparent diffusion coefficient. The parameter R is the ideal gas constant, and V_m is the molar volume of the precipitates. The parameter $C_\alpha(\infty)$ is the equilibrium solid-solubility of the precipitates in the matrix, and r_0 is the critical equivalent radius related to the temperature T and solute concentrations in the matrix. Only when $r > r_0$, $dr/dt > 0$, the precipitates keep growing. According to the size data collected from TEM,

the quadratic fitting curve of the particle ripening rate dr/dt and the reciprocal of the equivalent radius $1/r$ are shown in Fig. 5. The slope of the curves decreased as the Zr content increased. It means that the part “ $2DV_m C_\alpha(\infty)/RT$ ” in Eq. (1) decreased as Zr content increased. Under this experimental condition, “ $V_m C_\alpha(\infty)/RT$ ” was regarded as a constant. By calculation, the apparent diffusivity coefficients without Zr, 0.13 wt % Zr and 0.18 wt % Zr were D_0 , $0.92D_0$ and $0.75D_0$, respectively, as shown in Fig. 5. The addition of Zr hindered the precipitation evolution of Al-Mg-Si by reducing the apparent diffusivity coefficient of the Mg/Si atoms. This corresponded to the previously discussed results.

In Fig. 5, when the Zr content was large, the apparent diffusion rate of the Mg/Si elements was low. Zr decreased the growth rate of the Mg/Si particles. The inhibition of Mg and Si diffusion by the Zr content explained the low growth rate of the Mg/Si particles and also explained the improved heat-resistance of the Al-Mg-Si-Zr alloy. High strength, high conductivity, and high heat-resistant aluminum cables could be developed from the Al-Mg-Si-Zr alloys.

5. Conclusion

In summary, the Al-0.13 wt% Zr alloy showed the best performance under these experimental conditions. It had a conductivity of 34.44 MS/m, a tensile strength of 197 MPa, and an elongation of 17.8% after 72 h of aging, showing good heat resistance. The precipitation of the Mg/Si particles was influenced by the Zr element. The addition of Zr had a positive effect on the density of the Mg/Si particles and hindered the growth of the Mg/Si particles. Comparing the diffusion coefficients of the alloys with different Zr contents, when the Zr content was larger, the diffusion of Mg/Si atoms was more difficult.

Prime novelty statement

Our research analyzed how the Zr contents affect the mechanical

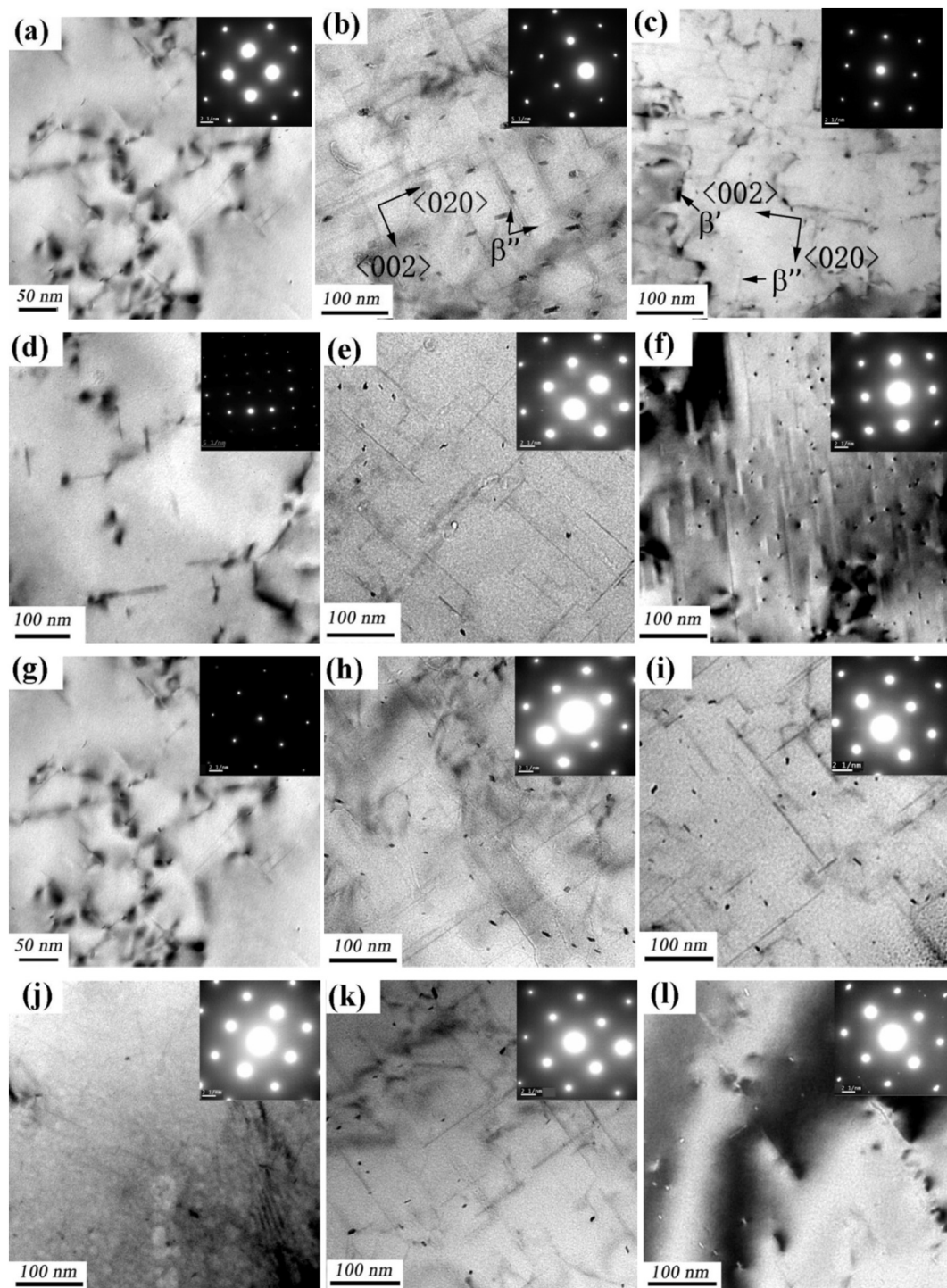


Fig. 3. TEM bright field images of alloys aged for different times at 200 °C: (a) 2 h of alloy without Zr, (b) 8 h of alloy without Zr, (c) 48 h of alloy without Zr, (d) 12 h of alloy 1, (e) 48 h of alloy 1, (f) 96 h of alloy 1, (g) 24 h of alloy 2, (h) 48 h of alloy 2, (i) 96 h of alloy 2, (j) 72 h of alloy 3, (k) 72 h of alloy 3, and (l) 96 h of alloy 3.

and electrical properties of Al-Mg-Si alloy cables during 96 hour aging. From mainstream view, three effects are attained by the addition of Zr: anti-recrystallization, strengthening and grain refining. We have come to view the issue in a different light. We found Zr addition can make the precipitation evolution of Mg/Si particles delay. The fundamental mechanism of the influence of the Zr additions on the Mg_2Si precipitates of the Al-Mg-Si alloy was discussed by means of quantitative transmission electron microscopy measurements. In particular, the diffusion coefficient of the Mg/Si atoms with different Zr contents was calculated.

Declaration of competing interest

There are no conflicts of interest.

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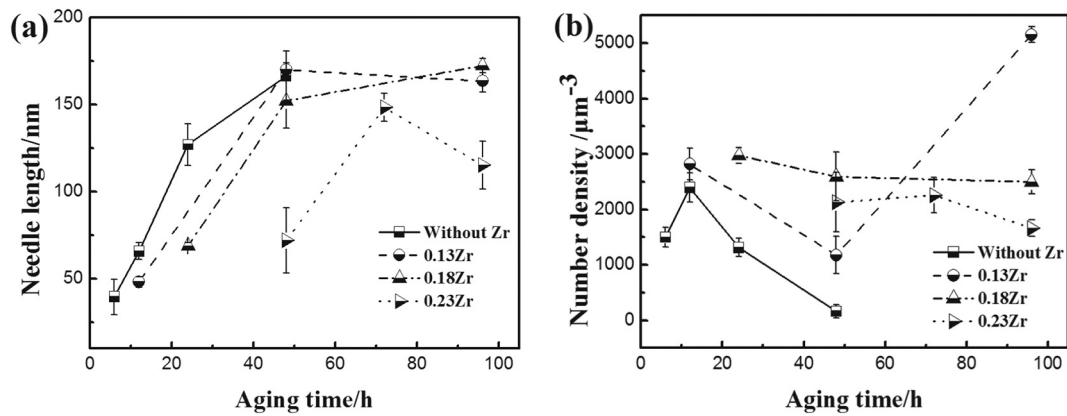


Fig. 4. Quantitative TEM measurements of the precipitates for all four alloys as a function of the artificial aging time: (a) average length of the precipitate needles and (b) precipitate number density.

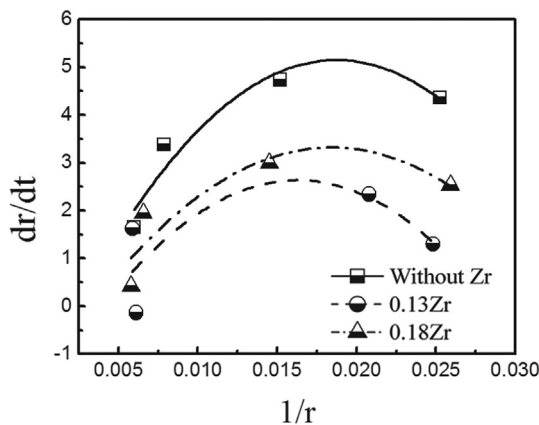


Fig. 5. Quadratic fitting curve of the particle ripening rate and the reciprocal of the radius for the tested alloys.

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